Scientific Report: Plant Acoustic Emissions

Experiment Start: 2026-06-22 18:55:01
Last Update: 2026-06-22 18:55:01
Duration: 0:00:00.078109
Session: 20260622_175202

Baseline Established: No
Baseline Duration: 0.0 minutes

METHODOLOGY

Statistical Methods:
• Significance Level: α = 0.05
• Drift Detection: ±2σ threshold
• Minimum Occurrences: 5 detections
• Frequency Binning: 1 kHz bins
• Trend Analysis: Linear regression

Tracking Metrics:
• Channel RMS and Peak amplitudes
• Persistent frequency peaks
• Frequency drift detection
• Soil moisture correlation
• Event rate analysis

SUMMARY STATISTICS

Channels Tracked: 0
Persistent Peaks: 202
Total Events: 264
Event Types: 5

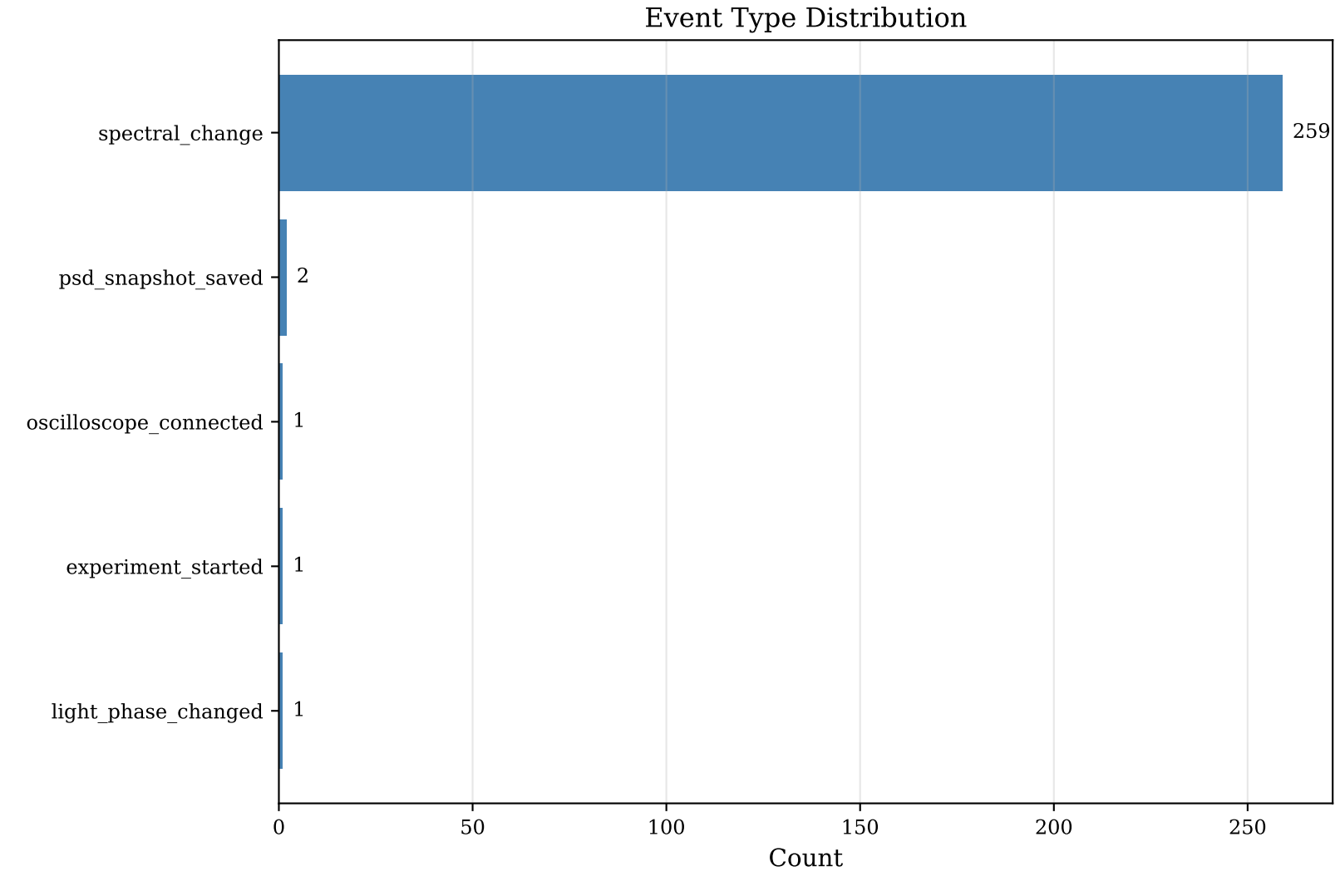
Significant Changes Detected: 47

Experimental Configuration

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OSCILLOSCOPE CONFIGURATION

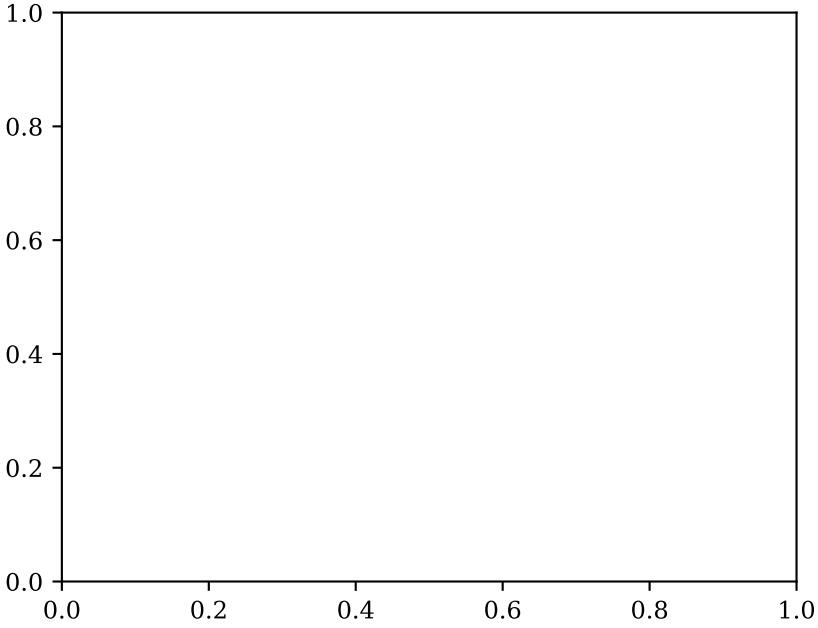
Sample Rate: 25 MSa/s
Memory Depth: AUTO
Time Base: 10 ms/div
Trigger Mode: EDGE
Trigger Source: CHAN1
Trigger Level: 20 mV

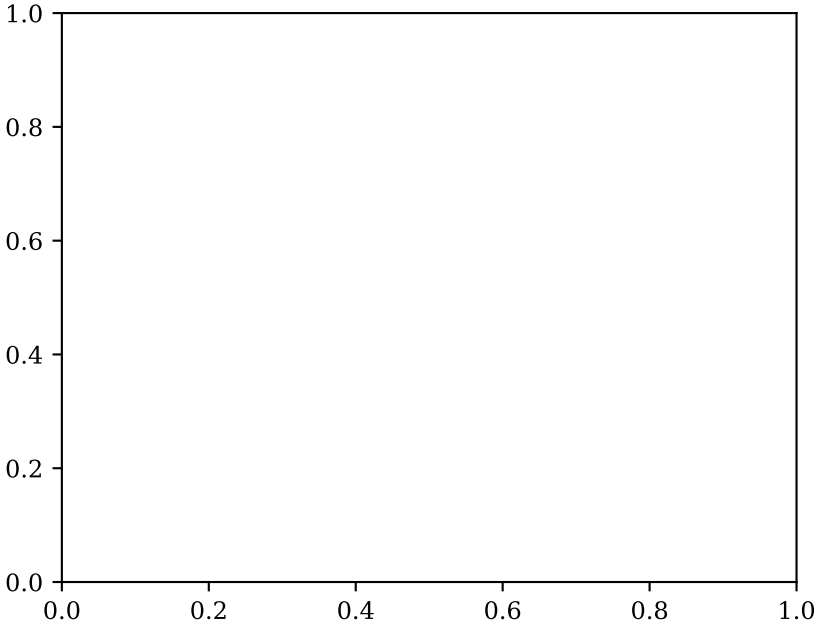
Channels:
• CH1: Piezo with LM358 amplifier (10:1 probe)
• CH2: Piezo with 820kΩ resistor (1:1 probe)
• CH3: Direct piezo (1:1 probe)
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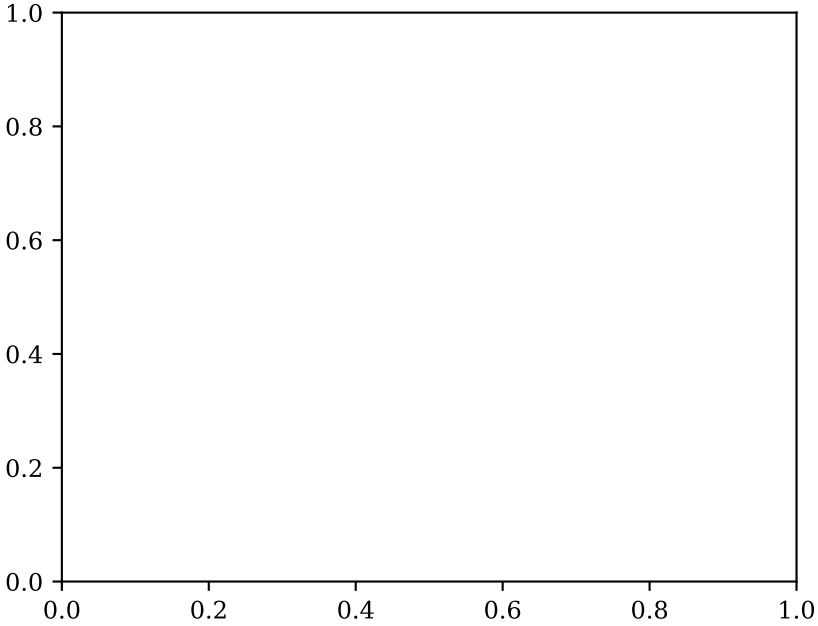


Channel Statistics

Channel Statistics Summary

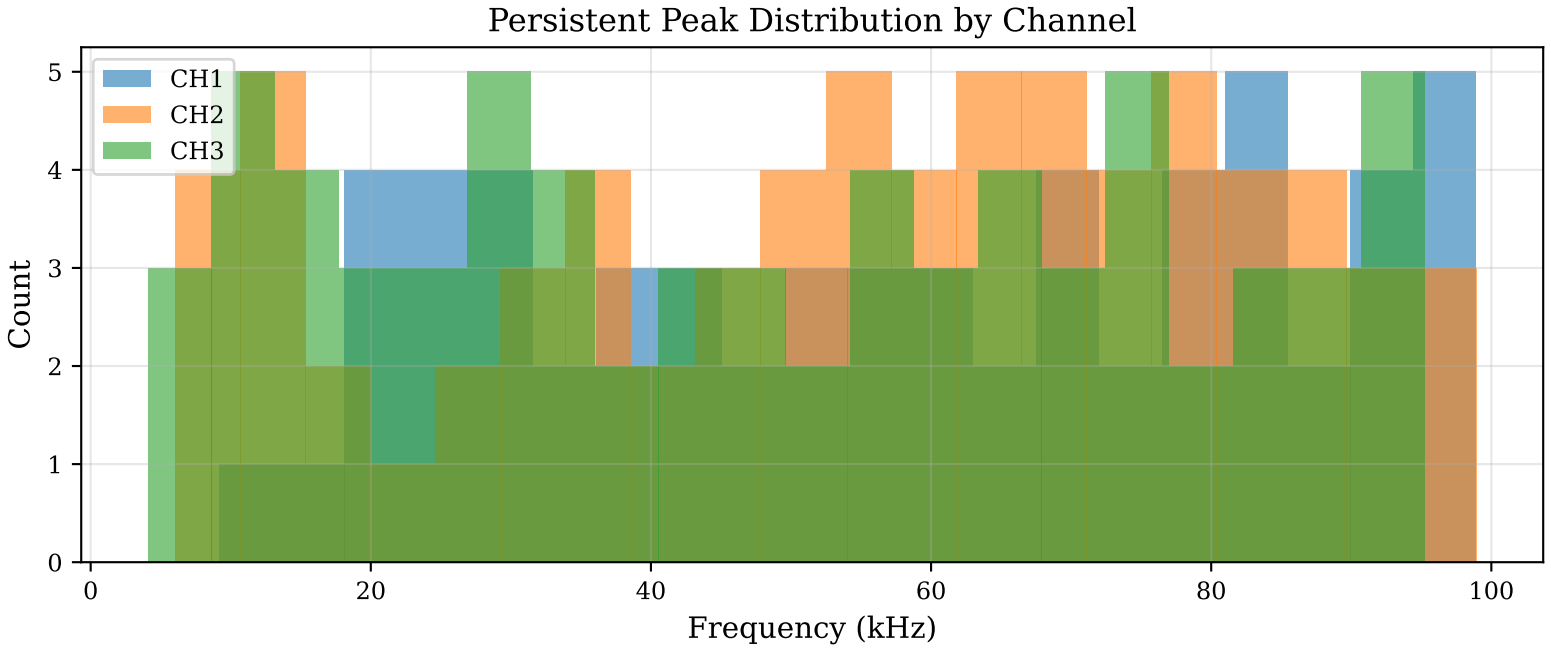


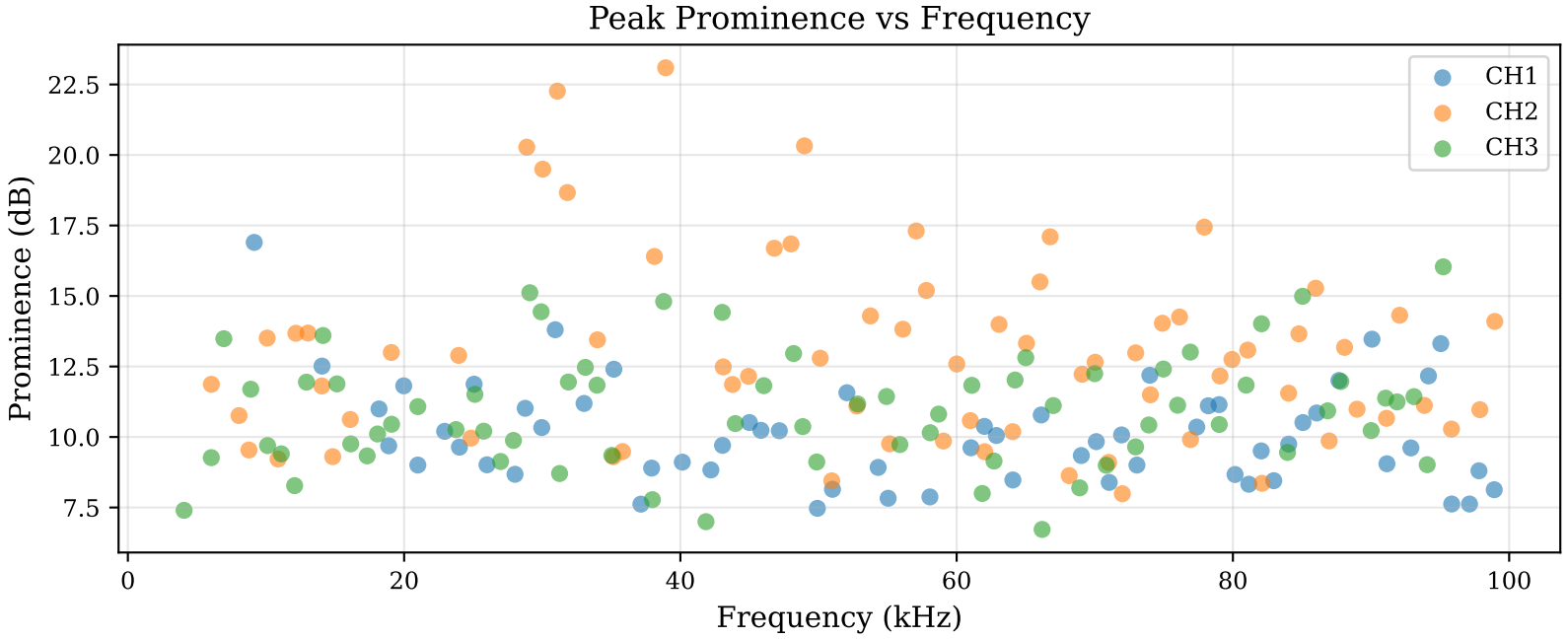




Channel	Mean	Std	Min	Max	N

Persistent Frequency Peaks

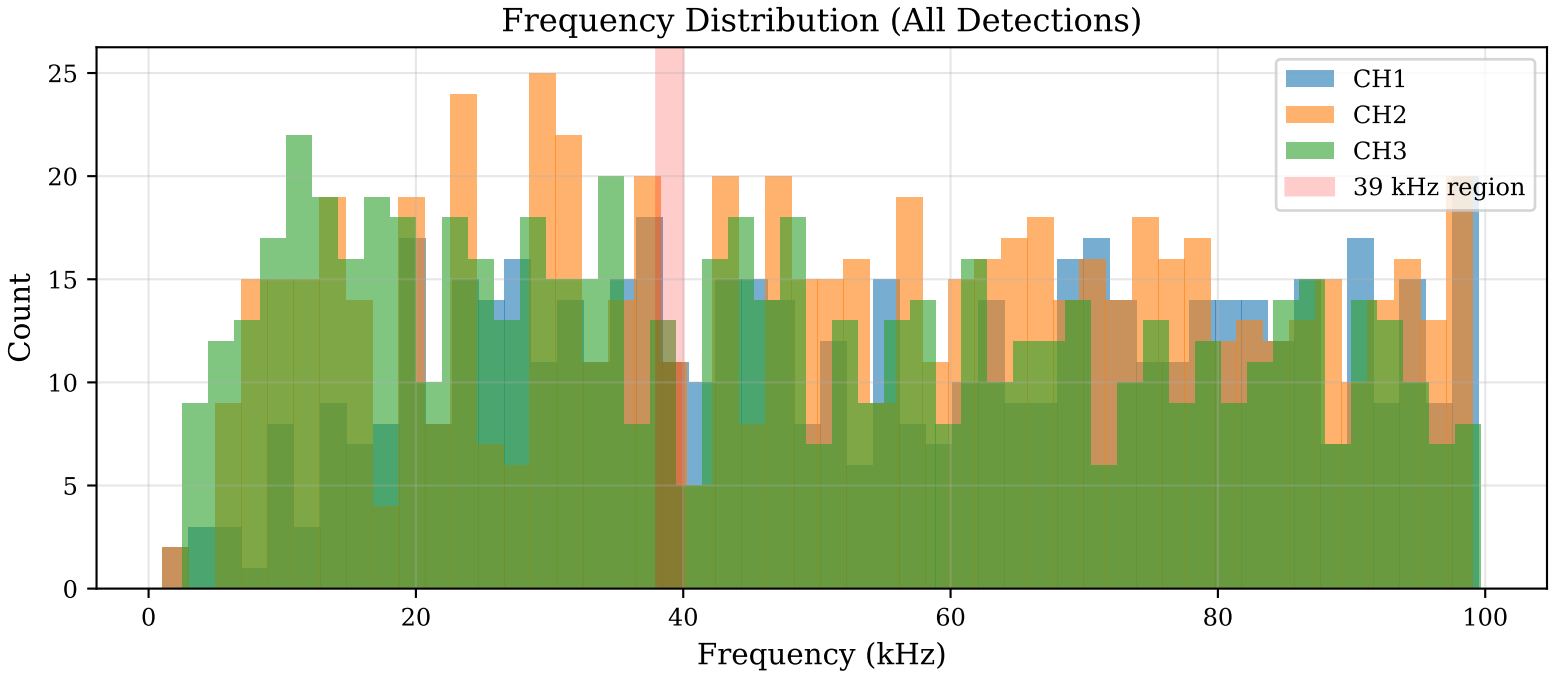


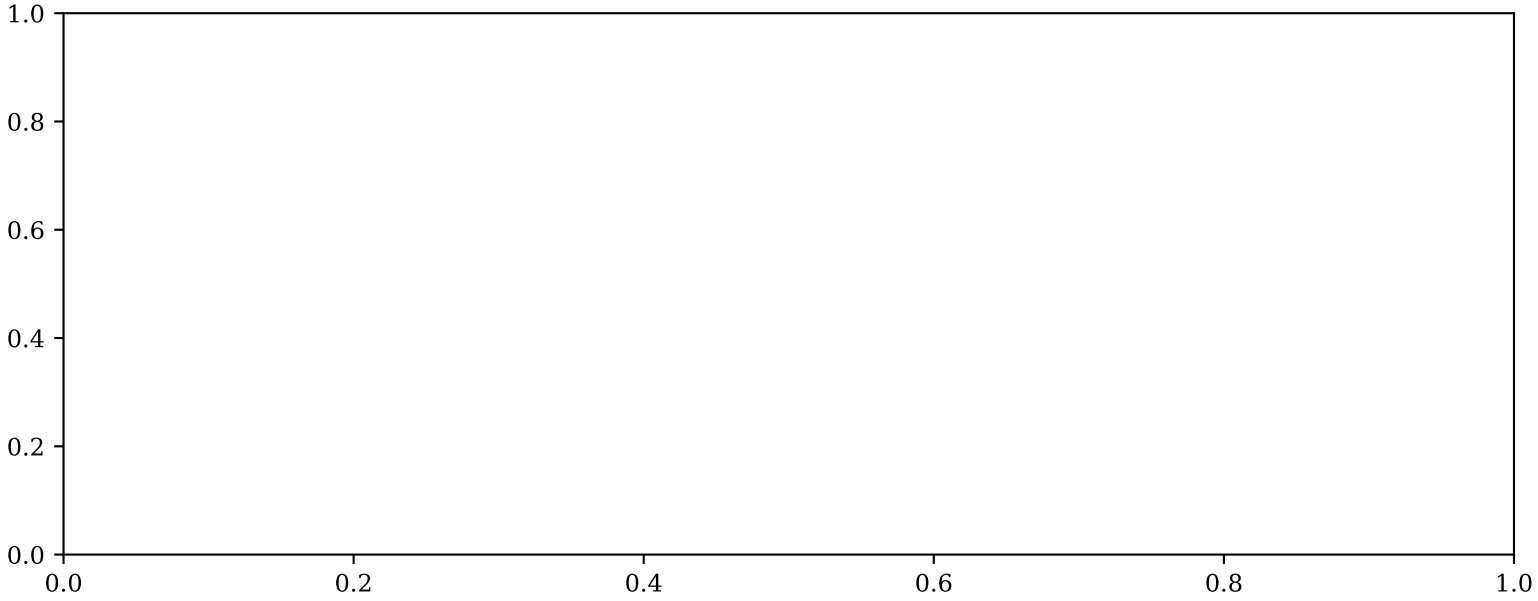


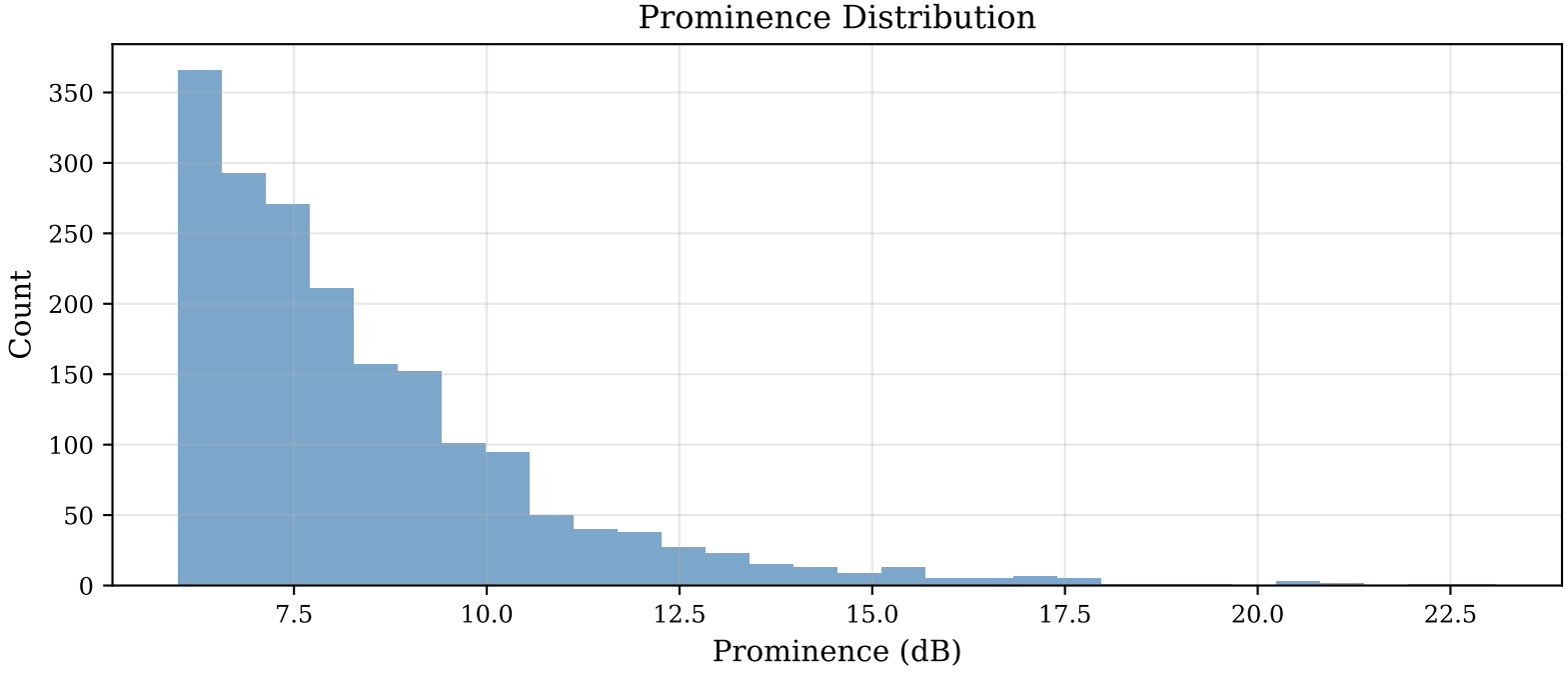
Top 10 Persistent Peaks by Prominence

CH	Freq (kHz)	Prom (dB)	Occ	Trend
CH2	38.9	23.1	5	drifting_down
CH2	31.1	22.3	15	drifting_down
CH2	49.0	20.3	5	drifting_up
CH2	28.9	20.3	17	drifting_down
CH2	30.0	19.5	7	stable
CH2	31.8	18.7	6	drifting_down
CH2	77.9	17.4	9	drifting_down
CH2	57.1	17.3	5	drifting_down
CH2	66.8	17.1	10	drifting_down
CH1	9.1	16.9	5	drifting_up

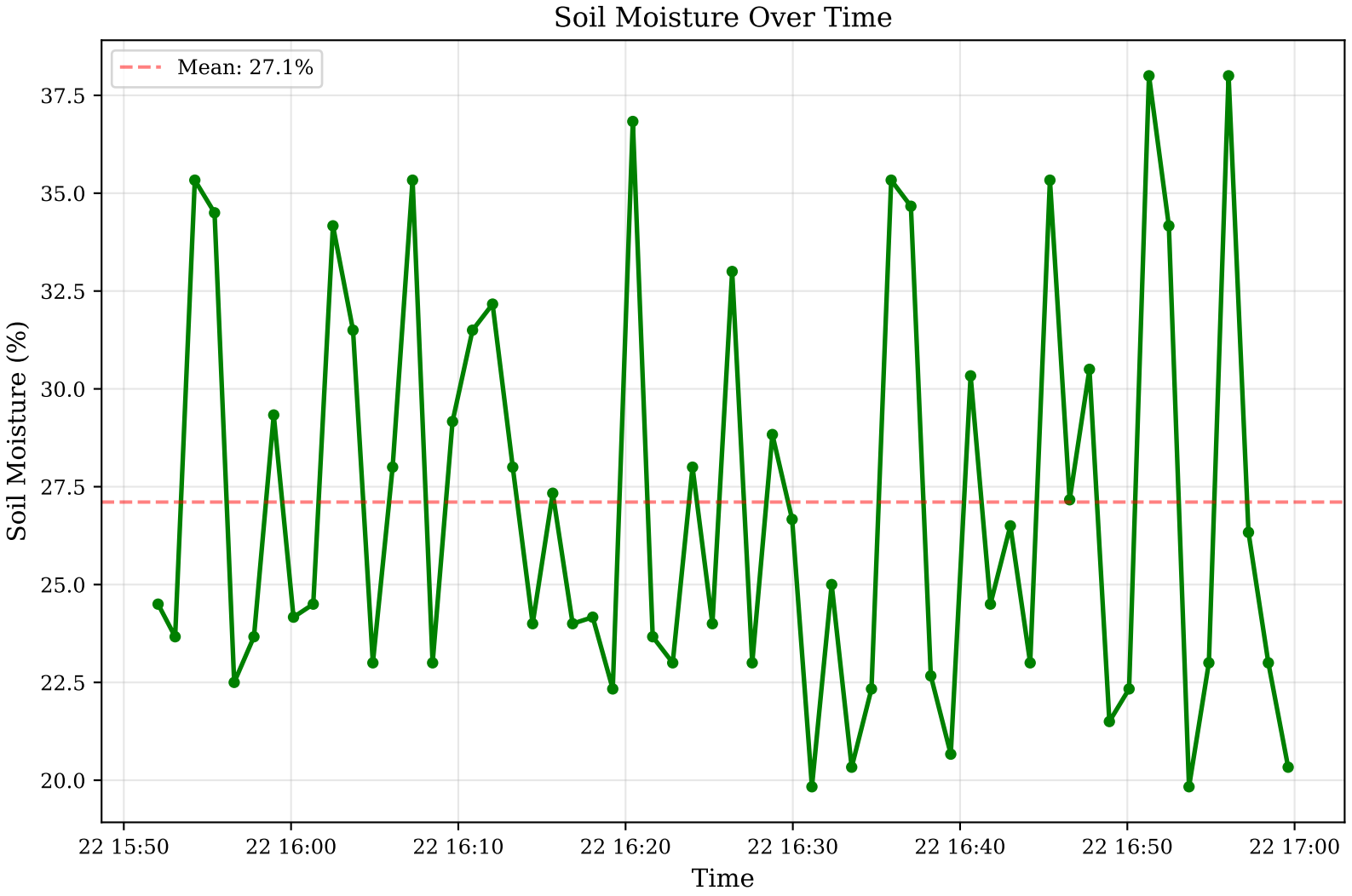
Frequency Distribution Analysis







Soil Moisture Analysis




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SOIL MOISTURE STATISTICS

Mean: 27.11%
Standard Deviation: 5.04%
Median: 24.75%
25th Percentile: 23.00%
75th Percentile: 31.25%
IQR: 8.25%
Min: 19.83%
Max: 38.00%
Samples: 54

INTERPRETATION

The soil moisture shows variable conditions
with a coefficient of variation of 18.6%.

Further analysis recommended.
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Summary and Conclusions

KEY FINDINGS

1. Persistent Frequency Peaks
   • Total persistent peaks: 202
   • CH1: 61 peaks
   • CH2: 72 peaks
   • CH3: 69 peaks

2. 39 kHz Region Analysis
   • Detections in 38-40 kHz range: 3
   • Channels affected: CH3, CH2, CH2
   • Mean prominence: 18.1 dB

3. Channel Activity
   • Most active channel: CH2
   • Least active channel: CH1

CONCLUSIONS

The experiment has identified 202 persistent frequency peaks across all three channels. The 39 kHz region shows some activity, particularly on CH2 and CH3.

RECOMMENDATIONS

• Continue monitoring for trend analysis
• Investigate 39 kHz activity for biological significance
• Correlate with environmental parameters
• Consider additional sensors for validation

STATISTICAL CONFIDENCE

• Significance level: α = 0.05
• Minimum occurrences: 5 detections
• Frequency resolution: 1 kHz bins
• Tracking duration: 0.0 minutes